



FEFM-YOLO11: Underwater Object Detection Algorithm Based on Improved YOLO11

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Abstract

Underwater object detection technology is extensively applied in subsea resource exploration and benthic environmental monitoring; however, the underwater environment presents significant challenges due to limited visibility caused by insufficient illumination and turbid water quality, severely impeding detection tasks. To address these challenges and enhance precision, this paper proposes FEFM-YOLO11—an enhanced YOLO11-based underwater object detection algorithm. Primarily, the algorithm substitutes two C3K2 modules in the backbone with a Feature Enhancement Module (FEM). Employing a multi-branch dilated convolutional structure, the FEM increases feature diversity, expands the network’s local receptive field, and enhances semantic representation of small objects, thereby strengthening detection capabilities. Subsequently, a Context-Aware Fusion Module (CAFM) is introduced in the Neck, which effectively models global and local features through integrated local feature capture (convolutional operations) and global feature extraction (attention mechanisms), consequently improving denoising performance. Furthermore, a dedicated small-target detection layer is incorporated to specifically boost detection performance for smaller underwater objects. Finally, the Wise IoU loss function was used for comprehensive evaluation and performance optimization, and the collaborative integration of these components effectively reduced the problem of missed detections caused by occlusion in dense cluster targets. Experimental results on the URPC2020 dataset demonstrate that the improved algorithm achieves enhancements of 2.0% points in mAP@50 and 1.9% points in mAP@50:95 compared to the baseline, confirming that FEFM-YOLO11 elevates detection precision and validates the feasibility and effectiveness of the proposed methodology.

Keywords YOLO11 · Object detection · FEM · CAFM · Loss function

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1 Introduction

With the rapid development of society, marine resources have garnered significant attention [1]. Due to the unique characteristics of underwater environments, such as insufficient lighting, turbid water, and debris interference, underwater object detection faces numerous challenges [2]. Specifically, factors like light attenuation and scattering cause blurred target edges and loss of texture details. These degradations exacerbate detection difficulties in densely distributed objects, where occlusion and low contrast increase false positives and missed detections. Notably, such optical interference is unique to underwater environments, unlike terrestrial high-density scenarios (e.g., crowd detection). Traditional object detection algorithms, such as sonar- or radar-based techniques, can partially address underwater detection tasks but often suffer from low accuracy and susceptibility to environmental interference.

In recent years, deep learning-based object detection algorithms have been widely adopted in underwater detection. These algorithms are broadly categorized into two types [3]: two-stage object detection algorithms (e.g., Fast R-CNN [4], Faster R-CNN [5], Mask R-CNN [6]), which typically achieve higher detection accuracy, and single-stage algorithms (e.g., SSD [7], YOLO series [8]), which exhibit faster inference speeds and fewer model parameters. Although two-stage algorithms achieve higher detection accuracy, their substantial computational overhead and inferior real-time performance make them unsuitable for underwater real-time detection requirements. In contrast, single-stage algorithms perform object classification and localization directly on feature maps without generating region proposals, thereby achieving faster detection speeds and better suitability for real-time tasks. In recent years, single-stage algorithms have gradually surpassed two-stage approaches, becoming the mainstream solution for underwater object detection research and applications.

Currently, researchers worldwide have conducted extensive studies in underwater object detection. For instance, Ye et al. [9] improved YOLOv5s using Varifocal Loss to address class imbalance and enhance detection accuracy. However, the modified model may over-emphasize hard-classified samples, leading to neglect of well-learned instances. Liang Xiuman et al. proposed an improved YOLOv7-based algorithm to tackle low visibility and color distortion in underwater environments by introducing a multi-information flow fusion attention mechanism (SGCA), which improved detection accuracy under blurred imaging conditions. Zhou Xin et al. [10] enhanced the YOLOv8 algorithm by integrating deformable convolutions (DCN) into the backbone network and designing an atrous spatial pyramid module (ASPF), significantly boosting underwater detection performance. Zhou et al. [11] developed the UODN network based on YOLOv8, incorporating a cross-stage multi-branch module (CSMB) and a large-kernel spatial pyramid module (LKSP), effectively addressing poor image quality and strengthening detection capability. Su Jia et al. [12] added a new detection layer to the neck network and adopted a decoupled head to improve small object detection.

Despite significant advancements in existing research, persistent challenges in underwater object detection indicate substantial room for further exploration. To address these limitations, this paper proposes FEFM-YOLO11 (Feature Enhanced and Fused Module-based YOLO11 algorithm for underwater object detection)—an enhanced architecture derived from YOLOv11—designed to mitigate feature degradation and missed detections caused by

densely distributed targets and image blurring in underwater environments while enhancing denoising performance. The algorithm's design originates from empirical observations of insufficient feature extraction in blurred underwater imagery and frequent overlap among densely clustered targets.

Inspired by multi-scale context modeling theory and dynamic feature fusion mechanisms, we construct a dual-module architecture. The first component is the Feature Enhancement Module (FEM) [13], which elevates detection capabilities through a multi-branch dilated convolutional structure. The second is the Convolution and Attention Fusion Module (CAFM), designed to capture both local and contextual information via convolution-to-attention coordination. The algorithm's nomenclature FEFM originates from this dual-module architecture, reflecting the technical evolution path from feature enhancement to adaptive fusion. The newly added 160×160 detection layer processes these shallow, high-resolution features, ensuring that faint signals from smaller targets are neither prematurely diluted nor overlooked. Furthermore, the adoption of the Wise-IoU loss function enhances training stability and bounding box localization accuracy through its dynamic focusing mechanism, effectively mitigating false positives and missed detections caused by localization errors. Contributions of this work are summarized as follows:

- To resolve the issue of insufficient feature extraction caused by the submergence of underwater target textural details within background noise, we introduce the Feature Enhancement Module (FEM). Its multi-branch dilated convolutional structure captures locally fine-grained textures and salient details, thereby enhancing feature extraction capabilities and ultimately elevating detection performance.
- We propose a feature enhancement-to-adaptive fusion design for degraded environments by incorporating the Convolution and Attention [14] Fusion Module (CAFM) [15] in the neck. CAFM achieves the optimal fusion of local details and global semantics, comprehensively modeling global and local features, effectively improving denoising performance.
- A dedicated 160×160 small-target detection layer is adopted to mitigate target information loss during feature degradation, preventing their signals from becoming undetectable in deep feature maps. This design reinforces shallow feature representation and captures local details, thereby improving object detection accuracy.
- To holistically optimize detection performance, we adopt the Wise-IoU [16] loss function. Conventional bounding box regression losses frequently exhibit violations of geometric constraints when processing ambiguous boundaries and irregular contours. Through its dynamic non-monotonic focusing mechanism, Wise-IoU adaptively modulates gradient assignments for anchor boxes, effectively reducing missed detections and false positives.

The remainder of this paper is organized as follows: Sect. 2 introduces the baseline YOLO11 architecture and its proposed enhancements. Sect. 3 presents experimental results and corresponding analysis. Finally, Sect. 4 provides primary conclusions and outlines directions for future work.

2 YOLO11 Network Model and Improvements

2.1 YOLO11 Network Model

YOLO11, the latest multi-task neural network [17] released by Ultralytics, integrates object detection, semantic segmentation, and image classification capabilities. Its fundamental architecture comprises three core components: a backbone network, a neck network, and detection heads. The backbone extracts hierarchical features from input images, the neck enhances multi-scale feature fusion, and the detection heads perform task-specific predictions. A detailed schematic of the network architecture is illustrated in Fig. 1.

YOLO11 significantly enhances feature extraction capabilities through an optimized backbone and neck architecture, achieving superior performance in object detection accuracy and complex task handling. Compared to YOLOv8 [18], YOLO11 replaces the CF2 module with the C3K2 module, introduces a novel C2PSA module after the SPPF block, and adopts YOLOv10's [19] head design philosophy by leveraging depthwise separable convolutions to reduce computational redundancy, thereby improving runtime efficiency.

The C3K2 module is a critical component in YOLO11's backbone network. Derived from the C2F module (designed for feature fusion), C3K2 introduces innovative modifications. Its behavior is controlled programmatically via the $c3k$ parameter. When $c3k = \text{False}$, the module functions as a standard C2F with conventional bottleneck structures. When $c3k = \text{True}$, the bottlenecks are replaced with C3 modules—specialized structures optimized for efficient feature extraction. This design enables adaptive feature extraction strategies, enhancing model flexibility and adaptability across diverse scenarios.

The C2PSA module, a core innovation in YOLO11, extends the C2F architecture by integrating a Point-wise Spatial Attention (PSA) mechanism. By embedding PSA blocks

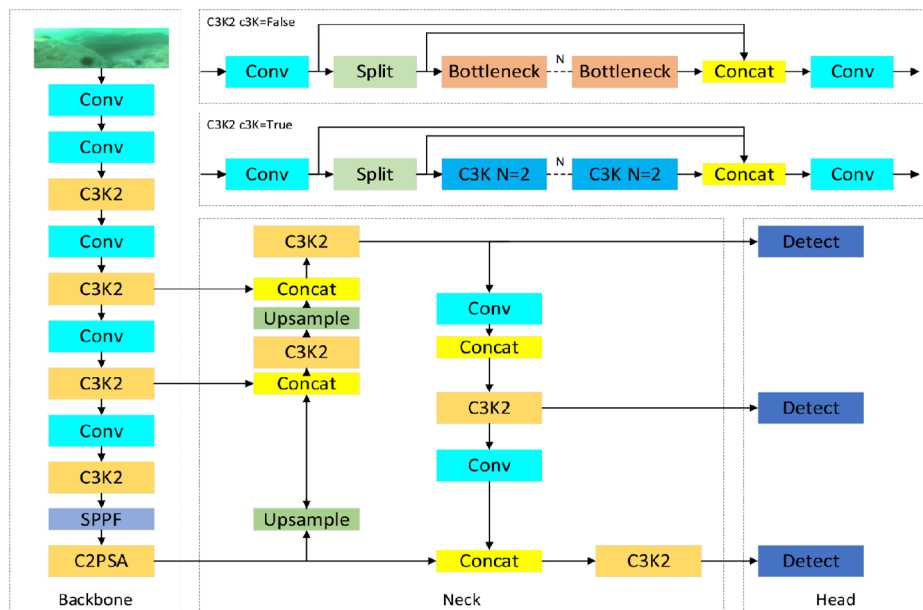


Fig. 1 Overall architecture of YOLO11

into the standard C2F framework, the module enhances spatial awareness of target details, enabling precise capture of critical features. The PSA block employs multi-head attention and feed-forward networks to dynamically focus on salient regions while suppressing irrelevant information. This design is particularly effective for complex image data, improving detection accuracy. Additionally, C2PSA supports optional residual connections (shortcuts) to optimize gradient flow and training stability, further boosting model performance.

YOLO11 retains the multi-scale prediction head design from earlier YOLO versions to accommodate objects of varying sizes. Leveraging feature maps generated by the backbone and neck networks, the head outputs detection boxes at three scales (low, medium, high). Specifically, predictions are derived from three feature layers (P3, P4, P5), corresponding to hierarchical detail levels:

- (1) P3: High-resolution features for small object detection.
- (2) P4: Intermediate features for medium-sized objects.
- (3) P5: Low-resolution features capturing global context for large objects.

This multi-scale architecture ensures comprehensive detection across object sizes.

2.2 Overall Architecture of FEFM-YOLO11

The overall architecture of the proposed FEFM-YOLO11 underwater object detection method is illustrated in Fig. 2.

Within the proposed architecture, we initially substitute select original backbone C3K2 modules with the Feature Enhancement Module (FEM). Adopting a multi branch dilated convolution structure to capture rich local features while enhancing the semantic informa-

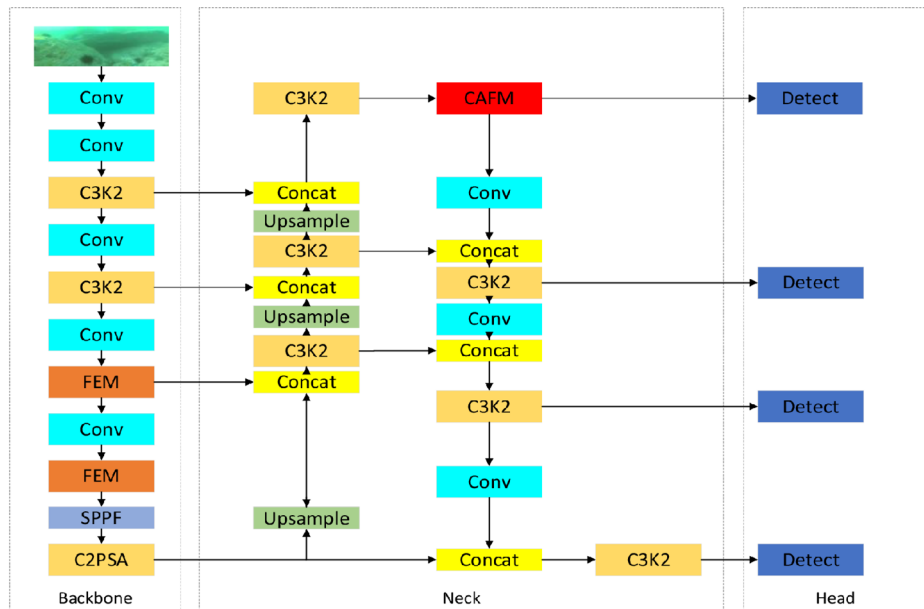


Fig. 2 Overall architecture of FEFM-YOLO11

tion of small objects. By expanding the receptive field, the FEM empowers the model to incorporate surrounding contextual information during small-object recognition. This architecture thus effectively augments the model's capability to distinguish targets from complex subsea backgrounds and noise [20].

Within the neck component, we introduce the Convolution and Attention Fusion Module (CAFM) to reinforce global and local feature extraction. This integrated design of FEM and CAFM modules enhances feature extraction capabilities while deepening contextual understanding, which facilitates the distinction of local features between adherent targets and improves denoising performance. Furthermore, a dedicated small-object detection layer is incorporated to process multi-scale feature maps more effectively, thereby boosting recognition performance for diminutive targets. Finally, leveraging a dynamic non-monotonic focusing mechanism, the Wise-IoU loss function is adopted to enhance bounding box regression precision.

2.3 FEM Module

In the backbone network of YOLO11, we replace the two bottom C3K2 modules with the FEM module to enhance feature extraction capabilities. The FEM consists of two components: standard convolution and dilated convolution [21]. Dilated convolution (also known as atrous convolution) expands the convolutional kernel's coverage by inserting spaces (zeros) between kernel elements. Its core parameter, the dilation rate, controls the spacing between kernel elements. When the dilation rate is 1, the dilated convolution behaves identically to a standard convolution. When the dilation rate is 2, the kernel elements are spaced one pixel apart, enlarging the receptive field and capturing multi-scale contextual information. The FEM module effectively improves the model's ability to extract multi-scale features, especially in scenarios that require fine-grained context awareness.

As shown in Fig. 3, the Feature Enhancement Module (FEM) adopts a multi-branch parallel architecture. It processes the input feature map in parallel using standard convolutions and dilated convolutions. Specifically, the input feature map is fed simultaneously into four branches: the first branch and the fourth branch perform 3×3 standard convolution and 1×1 convolution operations, respectively, to extract local detail features. The remaining two branches employ dilated convolutions to capture contextual information. The outputs from each branch are then fused via channel-wise concatenation (Concat) and output.

This fusion strategy significantly enhances feature diversity: the standard convolution preserves local texture details, while the added dilated convolution branches perceive horizontally elongated structures and vertical structural information of targets. The FEM achieves complementarity between local details and global context [22] through its multi-branch dilated convolutions. In the final step of the middle two branches, 3×3 convolutions with a dilation rate of 5 are used. This allows the convolutional kernel to cover a larger area (equivalent to an 11×11 effective receptive field) while maintaining the same parameter count. In contrast, the standard convolution has only a 3×3 receptive field. The expanded receptive field enables the network to perceive not only the target itself but also its surrounding context when detecting a small object. The contextual information obtained through the large receptive field can compensate for the lack of target specific information. Underwater targets are difficult to capture contextual information using traditional convolutions with fixed receptive fields due to their blurred edges and low resolution features. Our feature

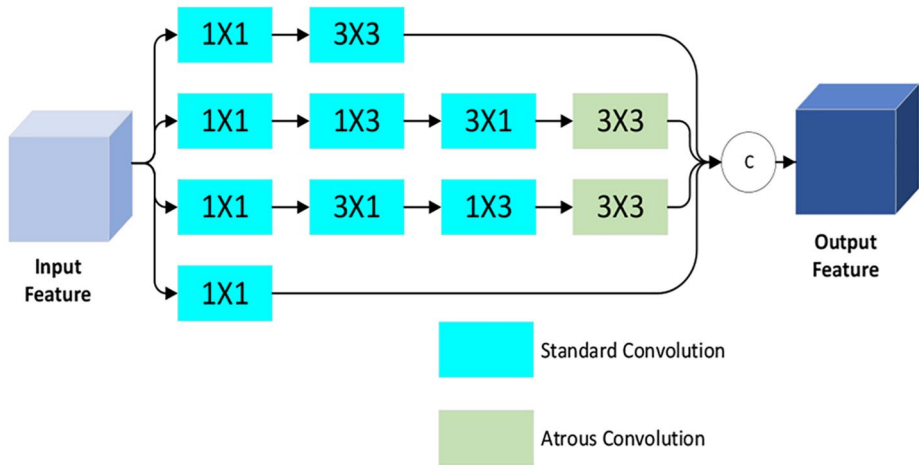


Fig. 3 Structure of the FEM module

enhancement module (FEM) solves this problem by expanding the effective receptive field to 11×11 through multi branch dilated convolution. Parallel branches with different dilation rates simultaneously capture local texture and contextual information. This utilizes environmental semantic clues to compensate for degraded targets, thereby improving recognition accuracy.

Despite significant image blurring and noise interference in underwater target imagery, this design enables the model to capture rich local features, enhances feature extraction capabilities, and thereby elevates target detection accuracy in challenging subsea environments.

2.4 CAFM Module

In the neck network of YOLO11, we introduce the Convolutional and Attention Fusion Module (CAFM). This module combines local feature capture (via convolutional operations) with global feature extraction (via attention mechanisms), enabling efficient modeling of both global and local image characteristics. As shown in the Fig. 4, the CAFM integrates these complementary capabilities to enhance detection performance, particularly in scenarios requiring fine-grained spatial awareness and contextual understanding.

The local branch in the figure involves the following key steps:

(1) Channel Dimension Adjustment:

First, the channel dimensions of the feature map are adjusted via a convolutional operation. This step modifies the number of channels while preserving the spatial dimensions (width and height) of the feature map, providing a more suitable representation for subsequent operations.

(2) Channel Shuffle:

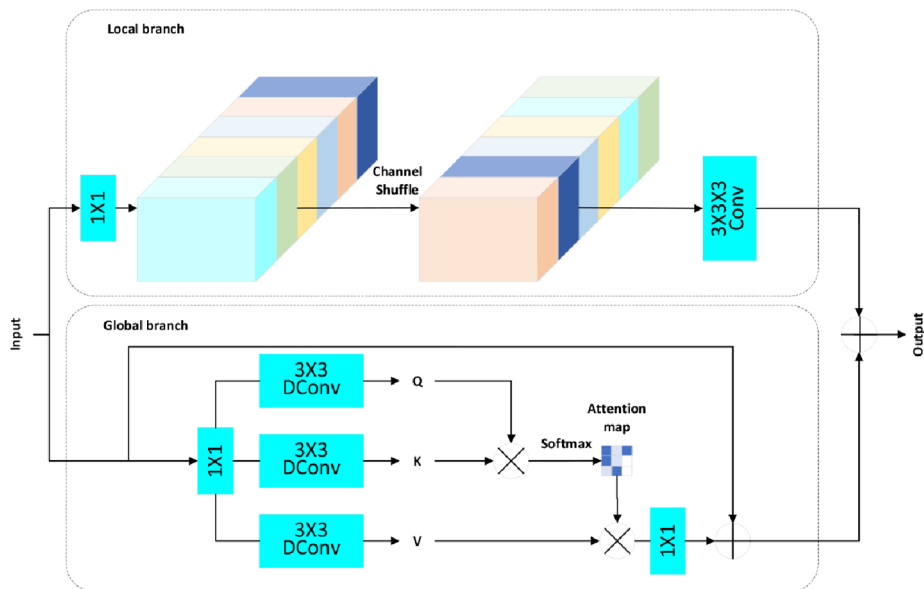


Fig. 4 Structure of the CAFM module

Next, a channel shuffle operation is applied. The input feature map is divided into multiple groups along the channel dimension. Within each group, depth-wise separable convolutions are used to enable cross-channel information exchange. Finally, the outputs of all groups are concatenated along the channel dimension to generate a new feature representation. This process enhances inter-channel feature fusion and interaction.

(3) Feature Extraction:

The final step employs convolutional operations to further refine feature extraction. These operations comprehensively capture local spatial and channel-wise features, improving the expressive power of the feature representation. The local branch can be summarized as:

$$F_{\text{conv}} = W_{3 \times 3 \times 3} (\text{CS} (W_{1 \times 1} (Y))) \quad (1)$$

In the equation, F_{conv} represents the output of the local branch, $W_{1 \times 1}$ denotes a 1×1 convolution, $W_{3 \times 3 \times 3}$ stands for a $3 \times 3 \times 3$ convolution, CS indicates the channel shuffle operation, and Y is the input feature. in Global Branch:

(1) Generating Query, Key, and Value:

We first generate Query (Q), Key (K), and Value (V) through a 1×1 convolution followed by a 3×3 depth-wise convolution. These tensors are reshaped into specific dimensions for efficient computation.

(2) Compute the attention map:

Reshape Q into $\hat{Q} \in \mathbb{R}^{\hat{H}\hat{W} \times \hat{C}}$ and K into $\hat{K} \in \mathbb{R}^{\hat{C} \times \hat{H}\hat{W}}$, then calculate the $A \in \mathbb{R}^{\hat{C} \times \hat{C}}$ attention map through the interaction between Q and K. This method reduces the computational burden by avoiding the calculation of the large $\mathbb{R}^{\hat{H}\hat{W} \times \hat{H}\hat{W}}$ -scale conventional attention map.

The output F_{att} of the global branch is defined as:

$$F_{att} = W_{1 \times 1} \text{Attention}(\hat{Q}, \hat{K}, \hat{V}) + Y \quad (2)$$

$$\text{Attention}(\hat{Q}, \hat{K}, \hat{V}) = \hat{V} \text{Softmax}(\hat{K}\hat{Q}/\alpha) \quad (3)$$

Where α is a learnable scaling parameter used to control the magnitude of the matrix multiplication between K and Q before applying the softmax function. Finally, the output of the CAFM module is computed as:

$$F_{out} = F_{att} + F_{conv} \quad (4)$$

To address the challenges of low contrast, high noise interference, and the overlapping/occlusion caused by densely distributed targets in underwater environments, we introduce the Convolutional and Attention Fusion Module (CAFM) in the Neck section. Its core principle lies in synergistically leveraging the local feature extraction capability of convolution and the global semantic modeling ability of attention mechanisms, thereby forming complementary advantages.

The local branch focuses on capturing fine-grained spatial features: After adjusting channel dimensions via a 1×1 convolution, a channel shuffle operation enforces information exchange across different channel groups, enhancing feature diversity. The critical $3 \times 3 \times 3$ convolution simultaneously extracts local features across both spatial and channel dimensions, exhibiting strong responses to detailed characteristics such as target textures, edges, and corner points. This approach enables the distinction of closely adjacent objects, facilitates the identification of minute gaps at contact points between two sea urchins or their discriminative texture patterns, and provides relatively stable local cues amidst noisy backgrounds.

The global branch focuses on contextual dependencies and importance weighting: After generating spatially structured Query/Key/Value (Q/K/V) representations via convolution, the attention mechanism computes global dependencies among all positions in the feature map. In low-contrast regions, it effectively suppresses responses from background noise unrelated to the target (even if they appear somewhat ambiguous under local convolution), significantly amplifying the semantic activation intensity of true target regions.

The fusion of local and global features achieves complementary advantages, ultimately producing outputs that integrate both fine-grained details and holistic semantic information. Local details serve as the foundation for recognition and localization, while global semantics guide the model to prioritize genuinely significant details and suppress distractions. Convolution provides a stable foundation of features, whereas attention dynamically adapts to scene content.

This design can effectively integrate global and local features, enhance contextual understanding, and improve denoising performance. The effective combination with FEM can reduce occlusion problems in densely clustered targets.

2.5 Small Target Detection Layer

For underwater target detection, although the initial YOLO11 model included three detection heads, the shallow layer of the network - with higher resolution - is more suitable for accurately capturing the position information of underwater targets. However, as the depth of the network increases, the feature information of underwater targets gradually decreases, resulting in insufficient utilization of smaller target details in shallow feature maps, ultimately reducing the performance of model detection.

To address the issue of scarce pixel information for small underwater targets, we have introduced a new high-resolution (160×160) small target detection layer next to the existing detection heads (80×80 , 40×40 , 20×20). The core mechanism of this design is to minimize the loss of small target feature information during downsampling. In the 80×80 detection layer, each feature point corresponds to an 8×8 region in the original image ($640/80=8$). In the 160×160 detection layer, each feature point corresponds only to a 4×4 region ($640/160=4$) in the original image. To prevent weak signals from being lost or diluted by background noise during subsequent downsampling and feature transformation processes. The added detection layer can improve the model's ability to capture fine information such as geometric shape and position. The target signal can be effectively extracted before being diluted by deeper network layers.

This design enhances shallow feature representation and captures local details, preserving more spatial detail information. Improved the accuracy of object detection.

2.6 Wise-IoU Loss Function

YOLO11 improves the model's ability to detect irregularly shaped targets and reduces the false positive rate (i.e. misclassifying the background as a target) by adopting the CIoU loss function. However, the CIoU loss function has limitations in measuring the aspect ratio difference between predicted bounding boxes and ground truth boxes, especially when dealing with underwater targets with blurred images, high noise interference, and dense targets. These limitations make it difficult to accurately capture shape changes, which may slow down the convergence speed during training and require more iterations to achieve optimal performance.

The traditional CIoU loss imposes excessive penalties on targets with fuzzy boundaries, resulting in inaccurate localization. At the same time, it uniformly processes all samples, making it difficult to distinguish sample quality and susceptible to interference from low-quality samples. To solve such problems. By using a dynamic non monotonic focusing mechanism, gradient resources are intelligently allocated to suppress the interference of low-quality and high-quality samples, and optimize the most effective "ordinary quality" samples for model improvement. This dynamic weighting reduces the adverse effects of fuzzy boundary targets on the loss function, while strengthening the learning of moderately difficult samples, ultimately improving the detection performance of the model, enhancing its generalization ability and overall performance [23].

Wise-IoU constructs distance-aware attention based on a distance metric, as shown in Eqs. 5 and 6, forming WIoUv1 with a dual-layer attention mechanism. Its primary role is to mitigate penalties imposed on low-quality samples due to geometric factors while improving the impact of these samples on the model's generalization capability. In the formu-

las, $\mathcal{R}_{WIoU} \in [1, e]$ is designed to significantly enhance $\mathcal{L}_{IoU}$ for anchor boxes of average quality, $\mathcal{L}_{IoU} \in [0, 1]$ is used to markedly reduce $\mathcal{R}_{WIoU}$ for anchor boxes of high quality. When the overlap between an anchor box (predicted box) and the ground truth box is high, the loss function reduces the emphasis on (i.e., decreases the weight of) the center point distance factor. This enables dynamic adjustment of attention to anchor boxes of varying quality levels.

The outlier degree (β) is defined to describe the quality of anchor boxes, where β is expressed as the ratio of $\mathcal{L}_{IoU}^*$ to $\mathcal{L}_{IoU}$ as shown in Eq. 7. Using β , a non-monotonic focusing coefficient r is constructed and applied to WIoUv1, resulting in WIoUv3 with a dynamic non-monotonic feature modulation (FM), as formulated in Eq. 8.

$$\mathcal{L}_{WIoUv1} = \mathcal{R}_{WIoU} \mathcal{L}_{IoU} \quad (5)$$

$$\mathcal{R}_{WIoU} = \exp \left(\frac{(x - x_{gt})^2 + (y - y_{gt})^2}{(W_g^2 + H_g^2)^*} \right) \quad (6)$$

$$\beta = \frac{\mathcal{L}_{IoU}^*}{\mathcal{L}_{IoU}} \in [0, +\infty] \quad (7)$$

$$\mathcal{L}_{WIoUv3} = r \mathcal{L}_{WIoUv1}, r = \frac{\beta}{\delta \alpha^{\beta - \delta}} \quad (8)$$

The introduction of WIoUv3 enables dynamic adjustment of attention levels for anchor boxes of different qualities during training. This approach reduces the competitiveness of high-quality anchor boxes while mitigating interference from low-quality samples, thereby enhancing the accuracy and stability of underwater target recognition.

3 Experimental Results and Analysis

3.1 Dataset

The dataset employed in this experiment is the URPC2020 dataset from the 2020 Underwater Target Detection Competition - Optical Track [24]. Collected from real underwater environments, this dataset includes four marine species: echinus, holothurian, starfish, and scallop, comprising a total of 5,543 images. The dataset was randomly partitioned into training, validation, and test sets at a ratio of 8:1:1. Example images of selected categories are illustrated in Fig. 5.

3.2 Experimental Environment

The experiments were conducted on the AutoDL platform using a server with the following specifications: Ubuntu 20.04 operating system, a 14 vCPU Intel® Xeon® Platinum 8362 CPU, an NVIDIA RTX 3090 GPU (24GB), and 45GB of RAM. The software environment was configured with CUDA 11.8, PyTorch 2.0.0, and Python 3.8. Training was performed for 300 epochs with an input image size of 640×640 pixels and a batch size of 16. The SGD

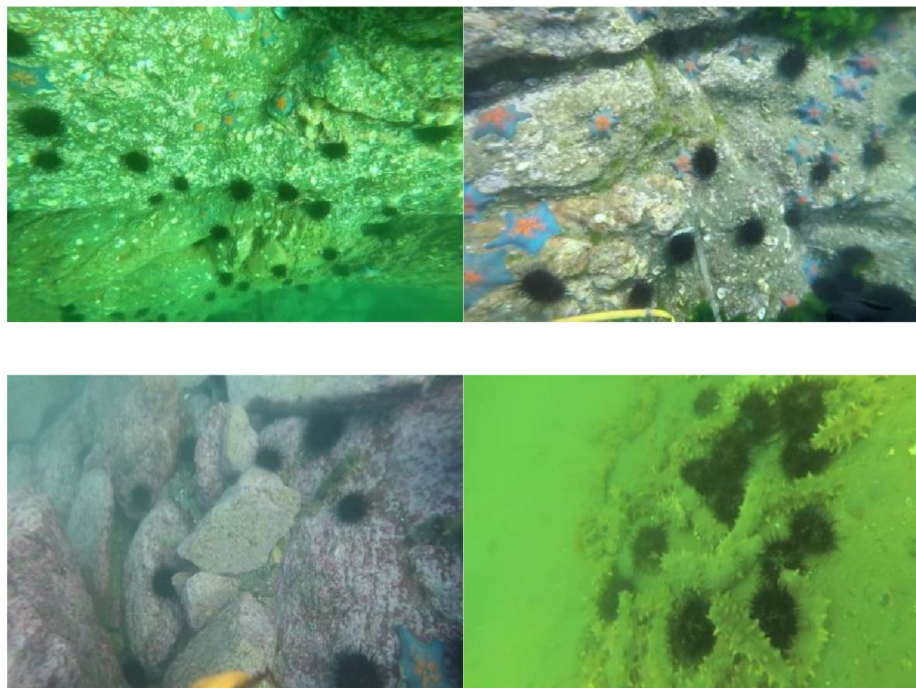


Fig. 5 Partial images from the dataset

optimizer was employed with an initial learning rate of 0.01, a weight decay coefficient of 0.0005, and momentum of 0.937.

3.3 Evaluation Metrics

In object detection tasks, Precision is a critical evaluation metric that measures the proportion of actual positive instances among those predicted as positive. The formula for Precision is defined as:

$$P = \frac{N_{TP}}{N_{TP} + N_{FP}} \quad (9)$$

Recall measures the proportion of all actually positive samples that are correctly predicted as positive by the model. The formula is as follows:

$$R = \frac{N_{TP}}{N_{TP} + N_{FN}} \quad (10)$$

mAP (mean Average Precision) is a widely used evaluation metric in object detection and information retrieval tasks, reflecting the model's average performance across all classes. In object detection, mAP is computed by averaging the area under the precision-recall curve (AP, Average Precision) for each class. The formula for mAP is defined as:

$$F_{mAP} = \frac{1}{N} \sum_{i=1}^N AP_i \quad (11)$$

3.4 Ablation Study

To validate the effectiveness of individual modules, we progressively integrated the proposed modules into the YOLO11 model and evaluated their impact using the URPC2020 dataset. The ablation study results are presented in Table 1, where F, C, E, and W denote the FEM module, CAFM module, additional detection head, and Wise-IoU loss function, respectively.

The ablation study results demonstrate the following:

This study employs YOLO11 as the baseline model and progressively incorporates the proposed improvements to validate the performance of each method in object detection. Table 1 presents the results of ablation studies conducted on the URPC2020 test dataset. Compared to the baseline model, our enhanced model demonstrates significant overall performance gains. Experiment 2 integrates the CAFM module, which enhances global-local feature modeling through a convolutional and attention fusion mechanism. Results show improvements of 0.6% in mAP50 and 0.2% in mAP50-95. To address small targets in the dataset, Experiment 3 introduces FEM feature enhancement to better capture local textures and contextual information. Results confirm the effectiveness of this improvement for small-object feature extraction, with mAP50 increased by 1.0% and mAP50-95 by 0.7% versus Experiment 1. Experiment 4 adds a small-target detection layer atop Experiment 3. Relative to the baseline (Experiment 1), this achieves +1.1% mAP50 and +1.2% mAP50-95, validating the layer's role in holistic accuracy improvement. Finally, Experiment 5 replaces the CIoU loss function with Wise-IoU in Experiment 4. Leveraging its dynamic non-monotonic focusing mechanism, Wise-IoU optimizes bounding box regression, reduces false detection caused by boundary ambiguity, and enhances detection stability. This culminates in a +2.0% mAP50 and +1.9% mAP50-95 gain over the baseline. Collectively, these ablation experiments confirm that our proposed algorithm substantially enhances overall performance, effectively meeting the demands of underwater object detection.

3.5 Comparison of Detection Results across Different Models

To validate the performance of the proposed algorithm, we conducted comparative experiments with state-of-the-art object detection algorithms under identical experimental settings and datasets. The selected benchmark tests include some of the latest models such as YOLOv5n, YOLOv6n, YOLOv8, and YOLOv10n. The results are summarized in Table 2.

Table 1 Ablation study results

Number	C	F	E	W	P/%	mAP50/%	mAP50–95/%	Params(M)
1	-	-	-	-	81.7	83.2	48.8	2.5
2	√	-	-	-	81.7	83.8	49	2.6
3	√	√	-	-	82.4	84.2	49.5	4.8
4	√	√	√	-	83.1	84.3	50	4.9
5	√	√	√	√	83.5	85.2	50.7	4.9

Table 2 Comparative detection results of different models on the URPC2020 dataset

Method	<i>P</i> /%	<i>R</i> /%	mAP50/%	mAP50–95/%
YOLOv5n	82.7	75.3	82.7	48.3
YOLOv6n	82.3	74.1	81.5	46.8
YOLOv8n	83.4	74.8	83.3	48.8
YOLOv10n	80.9	77.4	83.3	49
YOLO11n	81.7	77.1	83.2	48.8
SSD	74.2	68.7	75.4	38.8
Faster-RCNN	76.8	68.2	75.2	43.4
RetinaNet	70.9	64.8	76	42.6
CenterNet [25]	83.5	70.2	76.1	44.2
TOOD [26]	80.8	69.1	80.4	45.9
RTDETR-r18	82.5	69.3	81.4	46.5
FEFM-YOLO11n	83.5	77.8	85.2	50.7

Table 3 Comparative detection results of different models on the DUO dataset

Method	URPC		DUO	
	mAP50/%	mAP50–95/%	mAP50/%	mAP50–95/%
Faster-RCNN	75.2	43.4	75.9	54.8
SSD	75.4	38.8	79.7	50.7
YOLOv8n	83.3	48.8	81.7	61.8
YOLO11n	83.2	48.8	83.2	64.1
RTDETR	80.5	44.9	77.7	50.2
RetinaNet	76	42.6	78.2	64.4
FEFM-YOLO11n	85.2	50.7	85	65.4

From Table 2, it is evident that the proposed algorithm achieves significantly higher detection accuracy compared to other models. In addition, we also conducted experimental tests on the DUO [27] dataset. As shown in Table 3, the effectiveness of our method has been validated through comparison.

3.6 Visualisation Analysis

For better observation, Fig. 6 shows the PR curves of the baseline model (a) and our improved model (b). The PR curves demonstrate that our model achieves higher detection efficiency across various target categories. With a calculated mean Average Precision (mAP) of 85.2%, our model performed well.

Underwater, image blurring caused by turbid seawater and limited lighting makes detection considerably more challenging. Additionally, targets are often clustered and mutually occluded. To visually evaluate detection performance, we selected representative images containing smaller targets, densely clustered objects, and low-visibility blurred images for verification. As shown in Fig. 7, (a) represents the original model, while (b) denotes the enhanced FEFM-YOLO11 model. The comparative analysis of the target detection results shows that the improved YOLO11 network effectively improves the detection performance.

To verify whether our model reduces background noise around objects and reduces false positives, we used the HiResCAM heatmap visualization method. This method helps to accu-

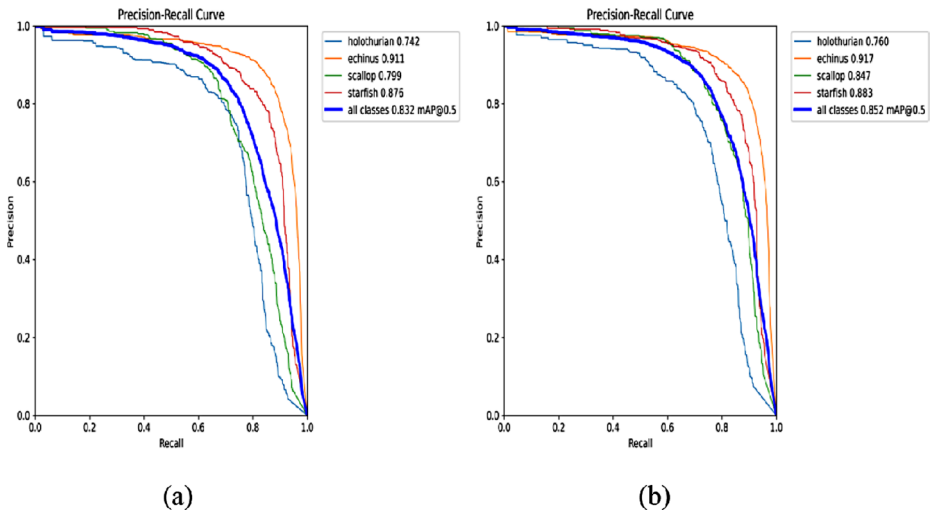


Fig. 6 PR curve

rately interpret the focal regions of deep learning models on images, highlighting the areas that the model is most interested in. This technology significantly improves the transparency and interpretability of the inference process in YOLO11 and FEFM-YOLO11. As shown in Fig. 8, comparative analysis indicates that FEFM-YOLO11 focuses more accurately on specific target areas. In the presence of background interference, FEFM-YOLO11 will pay more attention to the targets in the image.

4 Conclusions and Future Work

This paper proposes an improved algorithm based on the YOLO11 framework to enhance underwater target detection performance. To address challenges such as target blurring, occlusion, and small object sizes in underwater environments, the following optimizations were implemented:

- (1) **Backbone Enhancement:** Two C3K2 modules were replaced with FEM modules (Feature Enhancement Modules), which employ a multi-branch dilated convolutional structure to diversify feature representations and expand the network's local perceptual field, thereby improving semantic feature extraction for small targets.
- (2) **Neck Optimization:** The CAFM module (Convolutional Attention Feature Fusion Module) was introduced in the Neck, combining the local feature extraction capabilities of convolution with the global feature modeling of attention mechanisms. This enables effective fusion of global and local features, enhancing detection accuracy.
- (3) **Small Object Detection Layer:** An additional detection layer was incorporated to strengthen the model's capability to detect small-scale targets.
- (4) **Loss Function Improvement:** The original loss function was replaced with the Wise-IoU loss, which provides a more comprehensive evaluation of detection quality, significantly boosting performance.

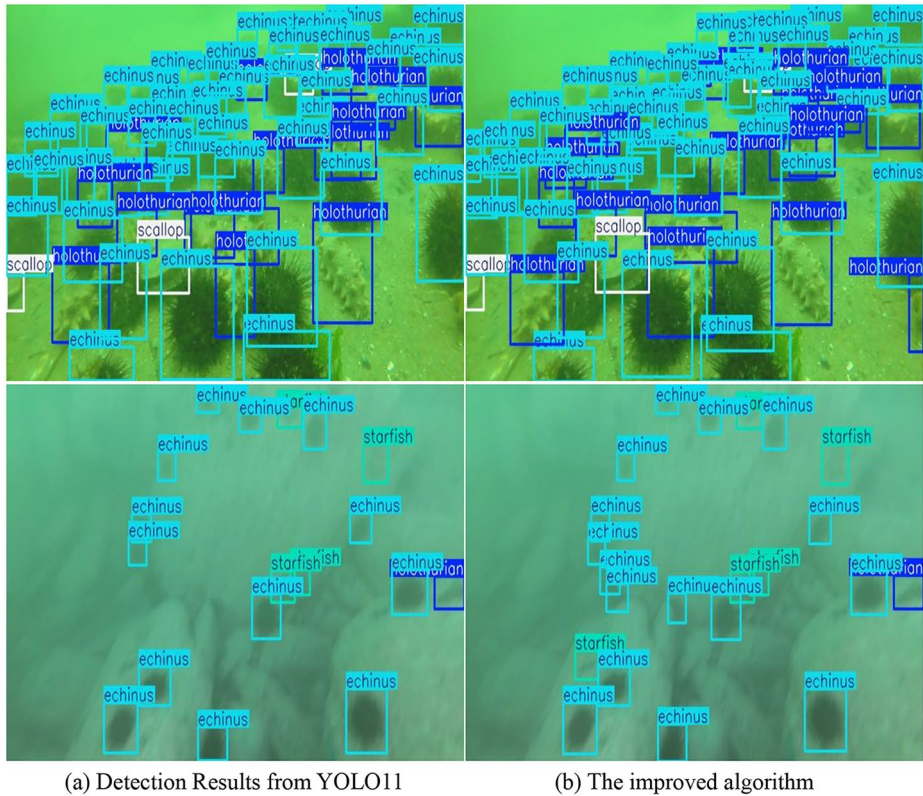


Fig. 7 Comparison of detection results

Experimental results demonstrate that the proposed algorithm achieves substantial improvements on the URPC2020 dataset. However, minor missed detection instances persist, indicating room for further refinement in underwater detection frameworks. Future research will focus on: Enhancing recognition efficiency in complex underwater scenarios and Exploring model compression techniques to deploy the algorithm on resource-constrained devices.

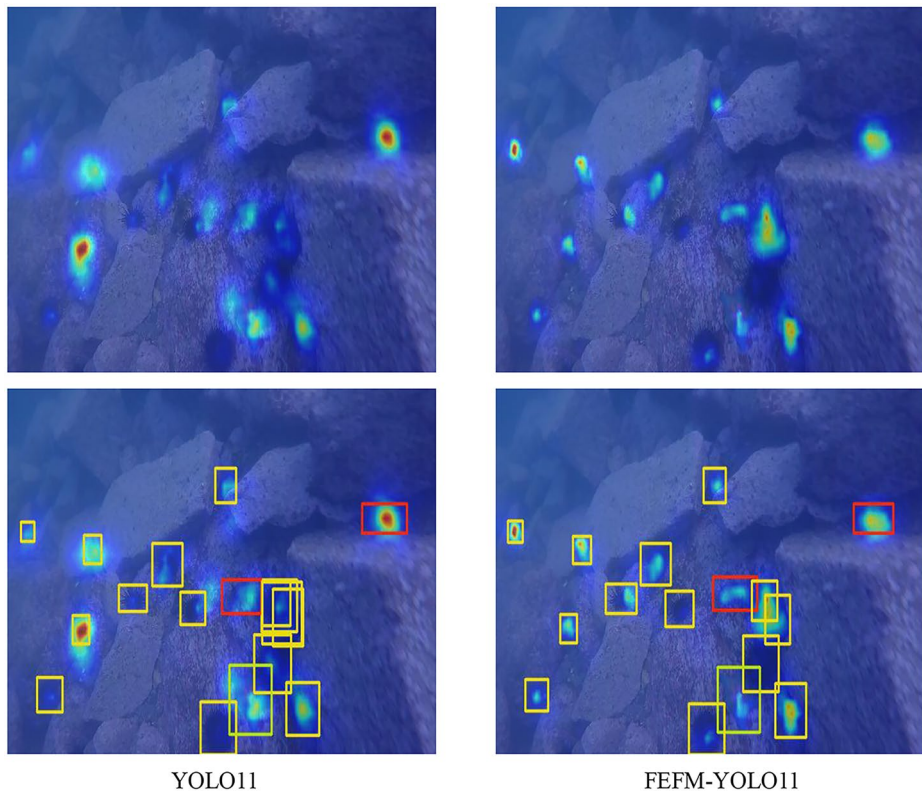


Fig. 8 Comparison of visualisations

Author Contributions Q.W. led the algorithm framework design and proposed the core improvement schemes for the FEM and CAFM modules; authored the main manuscript sections (Introduction, Methodology, Conclusions); designed the experimental protocol and supervised data validation; integrated research outcomes and coordinated manuscript revisions. Z.L. implemented the algorithm code and conducted model training; performed experimental data analysis and result visualization (Figs. 1, 2, 3, 4, 5, 6, 7 and 8); assisted in optimizing the Wise-IoU loss function; contributed to the writing and proofreading of the experimental section (Chapter 3). All authors reviewed and approved the final manuscript to ensure logical integrity and rigor of conclusions.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

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